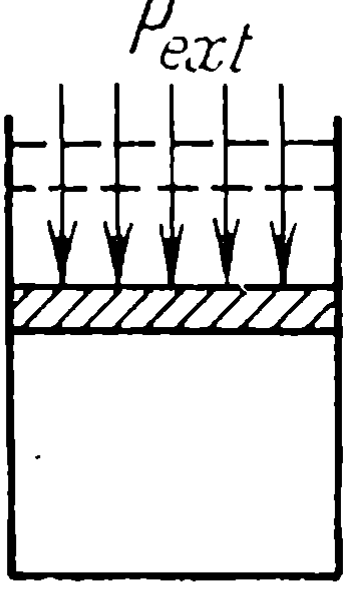

P_{ext}